

A Complete Elliptic Atlas for the Triaxial Euler Top: Periods, KKS Actions, and Fixed-Circle Obstructions

Route-A structural certificate C186

Abstract

For every strictly triaxial free rigid body, every nonzero angular-momentum sphere, and every energy, we give two explicit Jacobi charts, exact minimal periods, all axial stability rates, the intermediate-axis heteroclinic network, and KKS action-angle coordinates. The same atlas classifies every fixed component of a sampled time map. Resonant components are whole circles, so sufficiently large iterates do not have finite isolated fixed-point counts. The Hamiltonian flow has a canonical unitary Koopman owner but no intrinsic arithmetic gate, target divisor, or Hilbert–Pólya interpretation.

Keywords: Euler top; Jacobi elliptic functions; action-angle variables; Koopman group; clean fixed sets.

中文摘要

本文对任意严格三轴自由刚体、任意非零角动量球面及全部能量层，统一给出两套 Jacobi 椭圆图册、精确最小周期、六个主轴平衡点的稳定性、中间轴异宿分离网以及 KKS 作用量与角变量。固定时间映射在共振能量上固定整条圆，因此足够高的迭代不具有有限的孤立不动点计数。该系统虽有典范么正 Koopman 动力学，但没有内生素数时钟或目标除子。

关键词：欧拉陀螺；椭圆函数；作用量与角变量；固定圆；谱障碍。

1 Frozen reduced system

Let $0 < I_1 < I_2 < I_3$, set $a = I_1^{-1} > b = I_2^{-1} > c = I_3^{-1}$, and fix $|M| = G > 0$. For $m = M/G$ the Euler equation and normalized energy are

$$\dot{m} = Gm \times \text{diag}(a, b, c)m, \quad e = \frac{2E}{G^2} = am_1^2 + bm_2^2 + cm_3^2. \quad (1)$$

Hence $c \leq e \leq a$. Exact Jacobi representations are classical; we use the modern computational account in [1] and the explicit $4K$ convention in [2]. Our contribution is the all-energy theorem, consequence ledger, executable convention audit, and strict owner boundary, not priority for the classical solution.

2 All-energy theorem

Topology and axial modes. The endpoint levels $e = c, a$ consist respectively of $\pm e_3$ and $\pm e_1$. Every regular level on either side of b is two circles: $\text{sgn } m_3$ distinguishes $c < e < b$, while $\text{sgn } m_1$ distinguishes $b < e < a$. At $e = b$, the points $\pm e_2$ are joined by four heteroclinic branches. The squared nonzero tangent rates at axes 1, 2, 3 are

$$-G^2(a-b)(a-c), \quad G^2(a-b)(b-c), \quad -G^2(a-c)(b-c), \quad (2)$$

so the outer axes are elliptic and the intermediate axis is hyperbolic.

Low-energy chart. For $c < e < b$, put

$$A^2 = \frac{e-c}{a-c}, \quad B^2 = \frac{e-c}{b-c}, \quad C^2 = \frac{a-e}{a-c},$$

$$k^2 = \frac{(a-b)(e-c)}{(b-c)(a-e)}, \quad \Omega^2 = G^2(b-c)(a-e).$$

For $\sigma = \pm 1$ and $u = \sigma\Omega(t - t_0)$, every orbit is

$$(m_1, m_2, m_3) = (A \text{cn } u, B \text{sn } u, \sigma C \text{dn } u), \quad T_3(e) = \frac{4K(k)}{G\sqrt{(b-c)(a-e)}}. \quad (3)$$

High-energy chart. For $b < e < a$, put

$$A^2 = \frac{e-c}{a-c}, \quad B^2 = \frac{a-e}{a-b}, \quad C^2 = \frac{a-e}{a-c},$$

$$k^2 = \frac{(b-c)(a-e)}{(a-b)(e-c)}, \quad \Omega^2 = G^2(a-b)(e-c).$$

Then every orbit is

$$(m_1, m_2, m_3) = (\sigma A \text{dn } u, B \text{sn } u, C \text{cn } u), \quad u = \sigma\Omega(t - t_0), \quad T_1(e) = \frac{4K(k)}{G\sqrt{(a-b)(e-c)}}. \quad (4)$$

In both charts $0 < k < 1$. Although dn has real period $2K$, the vector contains sn and cn and has minimal period $4K$.

Separatrix and endpoint periods. At $e = b$, define

$$A_s^2 = \frac{b-c}{a-c}, \quad C_s^2 = \frac{a-b}{a-c}, \quad \rho = G\sqrt{(a-b)(b-c)}.$$

For $\varepsilon, \sigma = \pm 1$ the four branches are

$$m_1 = \varepsilon A_s \text{sech } u, \quad m_2 = \tanh u, \quad m_3 = \varepsilon \sigma C_s \text{sech } u, \quad u = \sigma \rho(t - t_0). \quad (5)$$

Thus they are heteroclinic from one intermediate-axis rotation to the other, not periodic orbits. Since $K(k) \rightarrow \infty$ as $k \rightarrow 1$, both regular periods diverge at $e = b$. At the stable endpoints,

$$T_3(c) = \frac{2\pi}{G\sqrt{(b-c)(a-c)}}, \quad T_1(a) = \frac{2\pi}{G\sqrt{(a-b)(a-c)}}. \quad (6)$$

3 Proof and action–angle atlas

Taking scalar products of $M \times I^{-1}M$ with M and $I^{-1}M$ proves both quadratic integrals. Lagrange multipliers on the sphere give the six critical points and the stated level topology; tangent linearization gives (2).

For (3), substitute $\text{cn}^2 = 1 - \text{sn}^2$ and $\text{dn}^2 = 1 - k^2 \text{sn}^2$. The constant and sn^2 coefficients in both quadratic integrals vanish. The rules

$$\text{sn}' = \text{cn dn}, \quad \text{cn}' = -\text{sn dn}, \quad \text{dn}' = -k^2 \text{sn cn}$$

reduce the three equations (1) to the displayed Ω^2 ; the proof of (4) is the same coefficient calculation with the dn component permuted. Taking $k \rightarrow 1$ and using $\text{sn}(u, 1) = \tanh u$ and $\text{cn}(u, 1) = \text{dn}(u, 1) = \text{sech } u$ proves (5). Taking $k \rightarrow 0$ proves (6).

Freeze $\{F, H\} = -M \cdot (\nabla F \times \nabla H)$ and $\tilde{F} = \{F, H\}$, so (1) has $\dot{M} = M \times \nabla H$. On a positive low component take $q = \arg(M_1 + iM_2)$ and $P_3 = G - M_3$; since G is a Casimir, $\{q, P_3\} = 1$. With $A_q = a \cos^2 q + b \sin^2 q$, its unsigned cap action $J_3 = (2\pi)^{-1} \oint P_3 dq$ is

$$J_3(e) = G \left[1 - \frac{1}{2\pi} \int_0^{2\pi} \sqrt{\frac{A_q - e}{A_q - c}} dq \right]. \quad (7)$$

For a positive high component take $q = \arg(M_2 + iM_3)$ and $P_1 = G - M_1$, so $\{q, P_1\} = 1$. With $B_q = b \cos^2 q + c \sin^2 q$, $J_1 = (2\pi)^{-1} \oint P_1 dq$ is

$$J_1(e) = G \left[1 - \frac{1}{2\pi} \int_0^{2\pi} \sqrt{\frac{e - B_q}{a - B_q}} dq \right]. \quad (8)$$

The negative components have the same unsigned action. Differentiating enclosed symplectic area gives the one-degree-of-freedom identity $|dJ/dE| = T/(2\pi)$. Finally $\theta = \pi u/(2K(k)) \pmod{2\pi}$ satisfies $\dot{\theta} = \pm 2\pi/T(e)$.

4 Sampled time, Koopman owner, and Route-A stop

Let Φ_τ be the time- τ map, $\tau > 0$. Its n th iterate fixes all six equilibria. A regular energy component is fixed pointwise exactly when

$$n\tau = qT(e) \quad (q \in \mathbb{Z}_{\geq 1}), \quad (9)$$

and no interior separatrix point is fixed. The periods are continuous, finite at the stable endpoints, and divergent at b . Therefore for every $\tau > 0$, all sufficiently large iterates have at least one fixed circle. The full-sphere isolated-cardinality Artin–Mazur coefficients are thus not finite; selecting one energy circle would change the frozen owner.

KKS area preservation makes $U_t f = f \circ \Phi_t$ a canonical unitary Koopman group. Off the measure-zero critical levels it is a direct integral over the two regimes and two components; circle Fourier mode ℓ carries the phase $\exp(2\pi i \ell t/T(e))$. This is a natural A4 coordinate. It does not supply rational primes, prime powers, a logarithmic clock, a target divisor, or a Hilbert–Pólya operator. Hence

$$(A0, A1, A2, A3, A4) = (\text{FAIL}, \text{WEAK}, \text{FAIL}, \text{FAIL}, \text{NATURAL QUANTIZATION}),$$

overall `ROUTE_A_REJECTED`, Route B false, under `NO_BAD_EULER_OR_ROOT_NUMBER`.

Exact certificate. The package records 180 rational sentinels across both regimes and 1,260 zero coefficient residuals. A producer-independent checker passes 4,268 assertions; a separate SymPy path proves 25 generic identities; replay is byte exact; 20 repaired-hash semantic corruptions and one stale-hash corruption are rejected. These rows audit conventions and do not replace the all-parameter proof above.

Revision-round focus. Round 2 makes the fixed-set statement iterate-exact and closes the positive-dimensional zeta obstruction, degenerate boundaries, evidence audit, and strict Route-A verdict.

Limitations. $G = 0$ is the stationary origin. Spherical and symmetric tops violate the strict inertia gaps and have equilibrium manifolds requiring separate elementary formulas. We do not continue the regular formula through $k = 1$, count a fixed circle as finitely many points, claim arithmetic semantics, or claim literature novelty or external review.

Declarations

Data and code availability. Exact evidence and deterministic code accompany this manuscript.

Ethics. No human, animal, clinical, personal, or sensitive data are used; approval is not applicable.

Contributions. This anonymous certificate records formal analysis, software, validation, drafting, and revision. AI systems are not authors.

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AI-use disclosure. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.

References

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- [2] E. G. Pina, “Drawing the free rigid body dynamics according to Jacobi,” arXiv:1505.06186 (2015).